GRE Prep

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1 Multivariable Calculus

Here, we include no proofs.

Exercise 1.1. Find the equation of the plane in $\mathbb{R}^3$ passing through $P = (0, 0, 1), Q = (1, 0, 0), R = (1, 1, 1)$.

Solution: We use a reference point and use the vector cross product:

$$v := \overrightarrow{PQ} = \langle 1, 0, -1 \rangle, \qquad w := \overrightarrow{PR} = \langle 1, 1, 0 \rangle$$

Now, the normal vector to P is $v \times w$:

$$v \times w = \langle 1, -1, 1 \rangle$$

The equation of the plane is given by

$$\langle 1, -1, 1\rangle \cdot \langle x-0, y-0, z-1\rangle,$$

which is simply $x - y + z = 1$. □

Exercise 1.2. Compute the arc length of the helix $\gamma(t) = (\sin(t), -\cos(t), t)$ from $t = 0$ to $t = 2\pi$.

Solution: We must use the arc-length formula $\int_t \|\gamma'(t)\| dt$.

$$\int_{t=0}^{2\pi} \sqrt{\cos^2(t) + \sin^2(t) + 1^2} dt = \int_{t=0}^{2\pi} \sqrt{2} dt = 2\pi\sqrt{2}.$$

□

The following is a nice way to determine if a limit doesn't exist:

Theorem 1.3. $\lim_{(x,y)\to(a,b)} f(x,y) = L$ *if and only if for all* $\gamma\colon [-,1,1] \to \mathbb{R}^2$ *such that* $\gamma(0) = (a,b)$, *then* $\lim_{t\to 0} f(\gamma(t)) = L$.

Exercise 1.4. Prove that $\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2}$ does not exist.

Solution: We consider the paths $\gamma(t) = (t,t)$ and $\delta(t) = (t,-t)$, which both tend to $(0,0)$ as $t \to 0$. Using γ, we have as follows:

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2} = \lim_{t\to 0} \frac{t^2}{t^2+t^2} = \frac{1}{2}$$

On the other hand, δ gives:

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2} = \lim_{t\to 0} \frac{-t^2}{t^2+t^2} = -\frac{1}{2}$$

Therefore, the limit doesn't exist. □

Exercise 1.5. Find $\lim_{(x,y)\to(0,0)} \frac{xy^2}{x^2+y^2}$.

Solution: We use polar coordinates and change the limit to $r \to 0$.

$$\lim_{(x,y)\to(0,0)} \frac{xy^2}{x^2+y^2} = \lim_{r\to 0} \frac{r^3 \cos\theta \sin^2\theta}{r^2} = \lim_{r\to 0} r\cos\theta\sin^2\theta$$

Now,

$$0 \le |\lim_{r\to 0} r\cos\theta\sin^2\theta| \le |\lim_{r\to 0} r| = 0,$$

so the limit is 0 via squeeze. □

1.1 Grad, Div, Curl

Recall that the gradient satisfies the following:

Proposition 1.6.

$$\nabla\phi = \left\langle \frac{\partial\phi}{\partial x}, \frac{\partial\phi}{\partial y}, \frac{\partial\phi}{\partial z} \right\rangle$$

Further, if we want the directional derivative of ϕ in the deriction of a unit vector $\hat{s}$, denoted $\frac{\partial\phi}{\partial s}$, we have

$$\frac{\partial\phi}{\partial s} = \hat{s} \cdot \nabla\phi.$$

Exercise 1.7. If $\phi = axy^2 + byz + cx^3z^2$, where a, b, c are constants, find $\nabla\phi$. Further, find the directional derivative of ϕ at the point $(1, 4, 2)$ in the direction towards the point $(2, 0 - 1)$.

Solution: First, we compute the gradient:

$$\nabla\phi = \left\langle ay^2 + 3cz^2z^2, 2axy + bz, by + 2cx^3z \right\rangle$$

Therefore, the gradient at $P = (1, 4, 2)$ is:

$$(\nabla\phi)_P = \langle 16a + 12c, 8a + 2b, 4b + 4c \rangle$$

Now, s is the difference between the two points, which is just $s = (2, 0, -1) - (1, 4, 2) = (1, -4, -3)$. Thus, $\hat{s}$ is

$$\hat{s} = \frac{\langle 1, -4, -3 \rangle}{\sqrt{26}}$$

Finally, the directional derivative $\frac{\partial\phi}{\partial s}$, is as follows:

$$\frac{\partial\phi}{\partial s} = (\nabla\phi \cdot \hat{s})_P = \frac{(16a + 12c) - 4(8a + 2b) - 3(4b + 4c)}{\sqrt{26}} = \frac{-16a - 20b}{\sqrt{26}}$$

□

Definition 1.8. Let F be a vector field.

- The *divergence* of F is the scalar product $\nabla \cdot F$, writing $\operatorname{div} F = \nabla \cdot F$.
- The *curl* of F is the vector product $\nabla \times F$, writing $\operatorname{curl} F = \nabla \times F$.

We recall some useful identities relating these different notions.

Theorem 1.9. *Let $\phi\colon \mathbb{R}^3 \to \mathbb{R}$ and F be a vector field (or more simply, a vector valued function).*

- $\operatorname{curl}(\nabla\phi) = 0$
- $\operatorname{div}(\operatorname{curl} F) = 0$
- $\operatorname{curl}(\operatorname{curl}(F)) = \nabla(\operatorname{div} F) - \nabla^2 F$

2 Complex Analysis

Here, we don't include proofs.

2.1 Basics of Holomorphic Functions

Theorem 2.1. *Let $f\colon \Omega \to \mathbb{C}$, where $\Omega \subseteq \mathbb{C}$ is open. The following statements are equivalent:*

- *f is holomorphic at $a \in \Omega$.*
- *f has a convergent power series $\sum_{n\geq 0} c_n(z-a)^n$ in an open disc centered at $a \in \Omega$.*

Definition 2.2. A complex-valued function is entire if it is holomorphic over $\mathbb{C}$.

Definition 2.3. Consider $f(z) = u + iv$ be a complex-valued function. Writing $z = x + iy$, the *Cauchy-Riemann equations* refer to the set of differential equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

Theorem 2.4. *Consider $f = u + iv\colon \Omega \to \mathbb{C}$, defomed on an open set $\Omega \subseteq \mathbb{C}$. If u and v are continuously differentiable and satisfy the Cauchy-Riemann equations on Ω, then f is holomorphic on Ω and $f'(z) = \frac{\partial f(z)}{\partial z}$.*

Example 2.5. $f(z) = \bar{z}$ is not holomorphic on $\mathbb{C}$. $\bar{z} = x - iy$, so $u'_x = 1 \neq -1 = v'_y$.

Exercise 2.6. Suppose that $f(x+iy) = u(x,y) + iv(x,y)$. If $u(x,y) = x^2 - y^2$ and $v(1,1) = 2$. Find $v(4,1)$.

Solution: The key is the FTC and the Cuachy-Riemann equations, because f is clearly holomorphic.

$$v(4,1) - v(1,1) = \int_1^4 v'_x(x,1)dx = \int_1^4 -u'_y(x,1)dy = \int_1^4 2dy = 6.$$

Therefore, $v(4,1) = 8$. Note the use of the Cauchy Riemann equations. □

2.2 Cauchy's Integral Formula

Theorem 2.7 (Cauchy)**.** *Let f be holomorphic in an open set Ω, then f has infinitely many complex derivatives in Ω. Moreover, for simple, closed, piecewise-smooth curve $\gamma \subset \Omega$ and any z lying inside γ we have*

$$\frac{d^n f(z)}{dz^n} = \frac{n!}{2\pi i} \oint_\gamma \frac{f(\eta)}{(\eta - z)^{n+1}} d\eta.$$

Exercise 2.8. Evaluate the following integral, for $\gamma = \{z\colon |z| = 2\}$:

$$\frac{1}{2\pi i} \oint_\gamma \frac{e^z}{(z+i)(z-i)} dz$$

Solution: We have as follows:

$$\frac{1}{2\pi i}\oint_\gamma \frac{e^z}{(z+i)(z-i)}dz = \frac{1}{2\pi i}\frac{1}{2i}\oint_{\{z:\,|z|=2\}}\left(\frac{e^z}{z+i}-\frac{e^z}{z-i}\right)dz = \frac{1}{2i}(e^i-e^{-i}) = \sin 1.$$

□

Theorem 2.9 (Morera)**.** *Suppose f is a continuous function in the open disc D such that for any triangle T contained in D*

$$\int_T f(z)dz = 0,$$

then f is holomorphic.

Corollary 2.10 (Liouville)**.** *A bounded entire function is constant.*

2.3 Residue Theory

Definition 2.11. Let $f(z) = \sum_{-\infty}^{\infty} a_n(z-z_0)^n$, for $0 < |z-z_0| < R$ be the Laurent series for f at z_0. The *residue* of f at z_0 is $\mathrm{Res}[f, z_0] = a_{-1}$.

Theorem 2.12. *Let f be holomorphic function inside and on a simple, closed, piecewise-smooth curve γ except at the singularities $z_1, \ldots, z_n$ in its interior. Then,*

$$\oint_\gamma f(z)dz = 2\pi i \sum_{j=1}^{n} \mathrm{Res}[f, z_j]$$

Remark 2.13. Note that we can find the residue of f at a pole z_0 of order m as follows:

$$\mathrm{Res}[f, z_0] = \lim_{z\to z_0} \frac{1}{(m-1)!}\frac{d^{m-1}}{dz^{m-1}}((z-z_0)^m f(z)).$$

Exercise 2.14. Evaluate $\oint_\gamma \frac{1}{z^5-z^3}dz$, where $\gamma = \{z:\,|z| = \frac{1}{2}\}$.

Solution: Let $f(z) = \frac{1}{z^5-z^3}$. The only relevant singularity in γ is $z = 0$. Thus, the integral is equal to $\mathrm{Res}[f, 0]$, which is a pole of order 3:

$$\begin{aligned}\lim_{z\to 0}\frac{1}{2!}\frac{d^2}{dz^2}\left(z^3\cdot f(z)\right) &= \frac{1}{2}\lim_{z\to 0}\left(\frac{-2z}{(z^2-1)^2}\right)' \\ &= \frac{1}{2}\lim_{z\to 0}\frac{-2(z^2-1)^2+4z(z^2-1)\cdot 2z}{(z^2-1)^4} \\ &= -1\end{aligned}$$

Thus, the integral evaluates to $-2\pi i$. □

3 Analysis and Point-Set Topology

We provide no proofs here.

3.1 Real Analysis

3.2 Topology

3.2.1 Open and Closed Sets

Lemma 3.1. *Let Y be a subspace of X. If U is open in Y and Y is open in X, then U is open in X.*

Remark 3.2. Note that if Y is not open in X, then there is no reason U should be open in X. As a counter example, consider $X = \mathbb{R}$ with the standard topology, $Y = [0, 1]$ (with the subpace topology), and $U = [0, \frac{1}{2})$. $U = (-\frac{1}{2}, \frac{1}{2}) \cap Y$, so is open in Y. However, clearly, U is not open in X, since opens in X are unions of open intervals.

Example 3.3. In the discrete topology on a set X, every subset of X is open. It follows that every set is closed as well.

Example 3.4. In any topology space X, $\emptyset$ and X are both closed and open.

A nice way to test if a set is closed in the subspace topology of some larger set is from the following:

Theorem 3.5. *Let $Y \subseteq X$. A is closed in Y if and only if it equals the intersection of a closed set of X with Y.*

It also follows then, from the lemma earlier, that:

Corollary 3.6. *Let $Y \subseteq X$. If A is closed in Y and Y is closed in X, then A is closed in X.*

There was also the notion of a closure, which is the intersection of all closed sets containing a set. Note that the resulting set is closed because arbitrary intersections of closed sets are closed.

Theorem 3.7. *Let $A \subseteq X$ be (only) a subset.*

- *$x \in \overline{A}$ if and only if every open set U containing x intersects A.*
- *If the topology of X is given by a basis, then $x \in \overline{A}$ if and only if every basis element containing x intersects A.*

Exercise 3.8. Let $X = \{n \in \mathbb{Z} \colon n \geq 2\}$ be the topoological space X with the topology generated by sets of the form $U_k := \{n \in X \colon n \mid k\}$, for $k \geq 2$, so that open sets of X are the empty set and arbitrary unions of the sets U_k. For any $n \in X$, determine the closure of the one point set $\{n\}$ in X.

Solution: Note that $m \in \overline{\{n\}}$ if and only if every open neighborhood of m meets $\{n\}$. In other words, we mean that if $m \in U_k$, then $\{n\} \cap U_k \neq \emptyset$, so $n \in U_k$. That is to say, if $m \mid k$, then we need $n \mid k$. Therefore, $m = ns$. Therefore, the closure is of the form $\{sn \colon s \in \mathbb{Z}_{>0}\}$. □

Definition 3.9. If A is a subset of the topological space X, then $x \in X$ is called a *limit point* of A if every neighborhood of x intersects A in some point other than x itself.

Remark 3.10. In other words, x is a limit point of A if x is in the closure of $A \setminus \{x\}$.

Theorem 3.11. *The closure of a set A in a topological space X is precisely the set A union with the set of limit points of A in X.*

3.2.2 Hausdorff

Recall the notion of Hausdorff:

Definition 3.12. A topological space X is *Hausdorff* if for each pair of distinct points $x, y \in X$, there are neighborhoods of x, y, respectively, which are disjoint.

Theorem 3.13. *Every finite point set in a Hausdorff space is closed.*

Theorem 3.14. *If X is a Hausdorff space, then a sequence of points of X converges to at most one point of X.*

Proposition 3.15. *Products and subspaces of Hausdorff spaces are Hausdorff.*

3.3 Miscellanious

4 Group Theory

4.1 Group Actions and Orbit-Stabilizers

Let G be a group acting on a set X.

Definition 4.1. The *stabilizer* of $x \in X$ of a group action G, G_x, is the set $\{g \in G \colon gx = x\}$. We denote the *orbit* of which contains x as $\mathcal{O}_x$.

Proposition 4.2. *G_x is a subgroup of G.*

Proof. Clear. □

Proposition 4.3. *Let X be finite. Then, $|\mathcal{O}_x| = [G : G_x]$.*

Proof. We define a bijective function between the cosets G/G_x and $\mathcal{O}_x$. We define $\phi \colon \mathcal{O}_x \to G/G_x$ as $y \mapsto gG_x$, where $gx = y$. Note that such a $g \in G$ exists because $y \in \mathcal{O}_x$. This is also well-defined in the sense that $\phi(y)$ is not dependent on g (easy check). The claim is that this is bijective, but this is, again, a direct check. □

Proposition 4.4 (The Class Equation)**.** *Let X be finite. Let $X_G = \{x \in X \colon gx = x, \forall g \in G\}$ be the fixed points of X. Then,*

$$|X| = |X_G| + \sum |\mathcal{O}_{x_i}|,$$

where the x_i are the representatives of each orbit.

Proof. This is clear. Elements not fixed by G are in some orbit, while those fixed by G are in X_G. □

Corollary 4.5. *Let G be a finite group and act on itself via conjugation. Then,*

$$|G| = |Z(G)| + \sum n_i,$$

where each n_i denotes the cardinality of each nontrivial conjugacy class. Further, n_i divides $|G|$ for all i.

Proof. The summation follows from the theorem above. The size of each orbit is also equal to the size of the cosets, which divide $|G|$ by Lagrange. □

Theorem 4.6. *Let $|G| = p^n$, for p prime. Then, G has nontrivial center.*

Proof. We let G act on itself via conjugation. Then, the previous proposition says

$$|G| = |Z(G)| + \sum n_i,$$

where each n_i is the size of the conjugacy classes. Then, $p \mid n_i$ and solution $p \mid |Z(G)|$. The identity is always in the center, so $|Z(G)| \geq 1$. Therefore, $|Z(G)| \geq p$ and so there is some nontrivial element in the center. □

Corollary 4.7. *Every group of order p^2 is abelian.*

Proof. By the previous theorem, if the center has order p^2, then clearly the group is abelian. Suppose not then. It follows that $Z(G)$ and $G/Z(G)$ both have order p, which are cyclic. Let $aZ(G)$ generate $G/Z(G)$. Then, we can write $gZ(G), hZ(G)$ as $g = a^m x, h = a^m y$ for some $x, y \in Z(G)$ and m, n. We just check for commuting of h and g, but this follow directly. □

Notation 4.8. We write $x \sim y$ if x and y belond to the same orbit.

Proposition 4.9. *If $x \sim y$, for $x, y \in X$, then G_x and G_y are iomorphic (as groups).*

Proof. Let $y = gx$, for some $g \in G$. Then, the map $\phi \colon G_x \to G_y$ defined as $a \mapsto gag^{-1}$ defines a bijective group homomorphism (thus, a group isomorphim). □

Theorem 4.10 (Burnide's Counting Lemma)**.** *Let k denote the number of orbit of X. We have as follows:*

$$k = \frac{1}{|G|} \cdot \sum_{g \in G} |X_g|.$$

Proof. The cardinality of elements of G which fix x is the sum $\sum_{g \in G} |X_g|$. Dually, viewed in terms of the stabilizer subgroups, this number is equal to the sum $\sum_{x \in X} |G_x|$. We also note that

$$\sum_{y \in \mathcal{O}_x} |G_y| = |\mathcal{O}_x| \cdot |G_x| = |G|.$$

Note that the first part follows by the previous lemma and the second equality follows by Lagrange. Now, we sum over the k distinct orbits to deduce that

$$\sum_{g \in G} |X_g| = \sum_{x \in X} X = k \cdot |G|.$$

□

4.2 Sylow's Theorems

Definition 4.11. A group G is a *p-group* if every element in G has as its order a power of p, for a prime p. A subgroup of a group is a *p-subgroup* if it is a p-group.

Recall Cauchy's theorem.

Theorem 4.12 (Cauchy)**.** *Let G be a finite group be a finite group with p dividing the order $|G|$. Then, G contains a subgroup of order p.*

Proof Sketch. Induct on the order of the group. The base case is immediate. Assume now that the result holds for all groups of order less than n (and p must divide n by hypothesis). We use the class equation and case work.

If the order of some conjugacy class is divisible by p, then we are done by the inductive hypotehsis.

Now, suppose that the order of no conjugacy class has order divisible by p. In this case, $[G : C_i]$ are all diviible by p (where C_i are the conjugacy classes), and the class equation implies that p divides the center. Now, $Z(G)$ is a finite abelian group, so the structure theorem implies that there is an element of order p in $Z(G)$. □

Corollary 4.13. *If G is finite, then G is a p-group if and only if $|G| = p^n$.*

The point of the Sylow theorem is to provide a partial converse to Cauchy's theorem.

Theorem 4.14 (Sylow I)**.** *Let G be finite and p be a prime such that p^r divides $|G|$. Then, G contains a subgroup of order p^r.*

Proof. Theorem 15.3 in [1]. □

Definition 4.15. A Sylow p-subgroup of a group G is a maximal p-subgroup of G.

Theorem 4.16 (Sylow II)**.** *Let G be a finite group and p be a prime dividing $|G|$. Then all Sylow p-subgroups of G are conjugate to each other.*

Proof. Theorem 15.6 in [1]. □

Theorem 4.17 (Sylow III)**.** *Let G be a finite group and let p be a prime dividing the order of G. Then the number of Sylow p-subgroups is congruent to* $1 \pmod{p}$ *and divides $|G|$.*

Definition 4.18. A *simple group* is one with no proper nontrivial normal subgroups.

Bibliography

[1] Thomas W. Judson and Stephen F. Judson. *Abstract Algebra: Theory and Applications*. Open-source textbook available at http://abstract.ups.edu/. Orthogonal Publishing L3C, 2020.